The board stays with you.

AI-generated concept visual; illustrative hardware.

Live blackboard lessons for weak connections.

CHALKLINE is a proposed teaching tool that uses on-device vision on a Snapdragon-powered HP PC to turn a physical chalkboard into a readable, versioned lesson stream. Students join from an ordinary phone browser and receive the teacher’s voice alongside changing board regions.

THE MOMENT WE WANT TO FIX

A teacher writes a minus sign, explains the next step, then erases the line. A remote student’s connection stalls at exactly the wrong moment. The lesson continues, but the reasoning is gone. CHALKLINE is designed to keep the original handwriting available, make earlier board states accessible, and recover the current board after a drop.

HOW THE LESSON WOULD FEEL

The teacher positions a USB camera, selects the board and starts teaching as usual. Local vision masks the presenter and tracks what is actually visible. The student listens, reads the last verified board, rewinds a step, and rejoins without silently receiving an obsolete image. Hidden or stale regions stay visibly marked; the system does not invent missing equations.

Built around the board.
Measured on the connection.

IMPLEMENTATION

A fixed camera feeds CPU-based board alignment and a candidate DeepLabV3+MobileNet segmentation model on the Snapdragon NPU, through ONNX Runtime QNN [1, 2]. CPU logic preserves observed pixels, confirms erasures across clear frames and produces ordered board patches. Compressed audio and checkpoints travel to a lightweight browser viewer. AI inference is local; remote delivery still needs connectivity.

THE CONTRIBUTION TO PROVE

The proposed contribution combines occlusion-aware board memory, explicit erasure handling and versioned recovery under a constrained link. Camera enhancement already exists: Teams supports board enhancement, but its documentation excludes blackboards [3]. Ordinary video already compresses changes between frames. CHALKLINE must therefore beat well-tuned video and periodic snapshots with audio on a fair test, rather than claim compression alone as novel.

64 kbps

Initial total per-receiver design target, including transport. Planned budget: 24 kbps audio + 32 board + 8 overhead. This allocation and all performance claims require validation.

VALIDATION & FIRST RELEASE

Build one fixed-camera, ten-minute lesson. Compare all three methods at 64/128/256 kbps, with writing, blocking, erasing and a ten-second network interruption. Measure blind symbol transcription, audio intelligibility, update and recovery delays, total bytes and actual CPU/NPU execution. Then seek a pilot with one teacher and 5–10 consenting adult learners. No benchmark, pilot or completed application is claimed at this stage.

PRIMARY REFERENCES

[1] Qualcomm AI Hub — candidate segmentation model

[2] ONNX Runtime — QNN execution provider

[3] Microsoft Teams — camera-based board enhancement

[4] Microsoft Learn — bandwidth guidance

Linked sources reviewed 8 September 2026. Generated imagery illustrates the proposed experience.